

UCH306: CHEMICAL ENGINEERING THERMODYNAMICS-I

L	T	P	Cr
3	0	0	3.5

Course Objective:

To understand the laws of thermodynamics, availability, the concept of entropy change, entropy generation and thermodynamic cycles as applied to a variety of engineering systems.

Introduction and Basic Concepts: Role of thermodynamics in engineering and science, Applications of Thermodynamics, Concept of Continuum, Macroscopic approach, Thermodynamics system and properties, Various processes, Thermodynamic equilibrium, Ideal gas, Vander Waals equation of state, Compressibility chart, Process: Flow and non-flow process, Cycle concept of work and heat, Specific heats, Zeroth law, Energy and its form, Pure substance, Thermodynamic diagrams, Triple point, Steam tables and their use.

First Law of Thermodynamics: Concept of internal energy & enthalpy, Energy equation as applied to a close and open system, PMMI, Transient flow processes.

Second Law of Thermodynamics & its Corollaries: Kelvin Plank and Clausius statements, Reversible and Irreversible processes, Carnot cycle, Clausius theorem and concept of entropy, Principle of increase of entropy, PMM2, Thermodynamic temperature scale, Second law analysis of control volume, Availability, Irreversibility, Availability function for open and closed system & second law efficiency.

Thermodynamic Cycles: Rankine cycle, Air standard cycles: Otto, Diesel, Dual and Brayton cycles.

Course Learning Outcomes (CLO):

Upon completion of the course, students will be able to:

1. estimate properties of pure substance using steam tables, property diagrams and equation of states.
2. analyze and solve problems involving closed system and open systems for both steady state and transient processes.
3. analyze the second law of thermodynamics for various systems and to evaluate the performance of heat engines, refrigerators and heat pumps.
4. analyze the performance of various power cycles and to identify methods for improving thermodynamic performance.

Text Books:

1. Rao, Y.V.C., *Thermodynamics*, Universities Press (2004).
2. Smith J. M. and Van Ness H. C., *Chemical Engineering Thermodynamics*, Tata McGraw-Hill (2007).

3. Nag, P.K., *Engineering Thermodynamics*, Tata McGraw Hill (2008) 3rd ed.

Reference Books:

1. Cengel, Y. A. and Boles, M., *Thermodynamics: An Engineering Approach*, Tata McGraw Hill (2008).
2. Sonntag, R.E., Borgnakke, C. and Van Wylen, G.J., *Fundamentals of Thermodynamics*, John Wiley (2007) 6th ed.
3. Rogers, G. and Mayhew, Y., *Engineering Thermodynamics*, Pearson Education (2007) 4th ed.

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessional (may include Quizzes/Assignments evaluations)	25